

PROJECT: DC MOTOR SPEED CONTROL

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Report Title	DC MOTOR SPEED CONTROL
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1. Introduction

The purpose of this project was to design, wire, program, and test a complete DC motor speed control system using an Allen Bradley PLC, industrial DC drive, HMI, and SCADA platform.

The project demonstrates how modern automation systems control motor speed using analog outputs, monitor real-time feedback using sensors, and provide user interaction through graphical interfaces.

The DC motor speed is controlled by a **0–10 V analog signal** sent from the PLC to the DC drive. The operator can enter speed, ramp up time, and ramp down time through the HMI or SCADA system. The PLC processes these values and controls the motor accordingly.

Actual motor speed is measured using an incremental encoder mounted on the motor shaft. Encoder pulses are counted by the PLC and converted into RPM.

This project integrates industrial concepts including:

- PLC programming
- Analog scaling
- Motor drives
- HMI development
- SCADA monitoring
- Encoder feedback
- Industrial wiring and troubleshooting

2. System Components

Hardware Used

- Allen Bradley ControlLogix PLC
- Digital Input Module
- Digital Output Module
- Analog Output Module
- DC Drive Board
- 24 V DC Motor
- Incremental Encoder
- Start Pushbutton
- Stop Pushbutton
- Indicator Lamps
- HMI PanelView Terminal

Software Used

- Studio 5000 Logix Designer
- FactoryTalk View ME
- FactoryTalk View SE / SCADA

3. Electrical Wiring and Hardware Connections

The system was wired using PLC input/output modules, DC drive board, motor, encoder, and indicator lamps.

Wiring Functions

Digital Inputs

Used for pushbuttons:

- **AT_START_PB** → Start command
- **AT_STOP_PB** → Stop command

Digital Outputs

Used for status lamps:

- Running Light
- Stopped Light
- Accelerating Light
- Decelerating Light

Analog Output

Used to send speed reference voltage:

- **VOUT-0 = 0 to 10 V**

Connected to:

- Drive WP terminal
- Return to LO terminal

DC Drive Connections

The drive receives:

- 24 VDC supply
- Analog speed reference from PLC

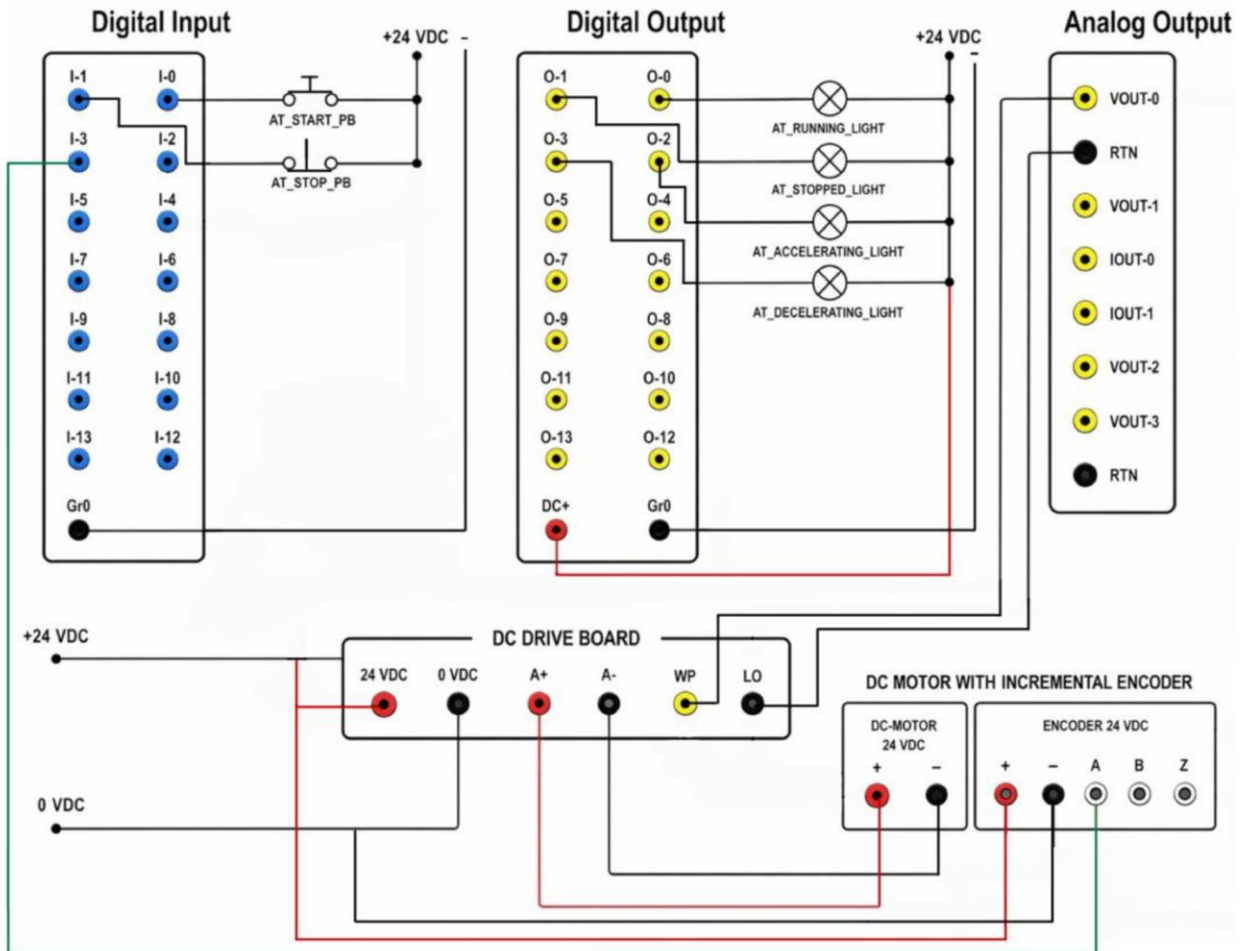
Outputs:

- Armature voltage to motor terminals A+ and A-

Encoder Connections

Encoder powered by 24 VDC and pulse channel connected to PLC input.

Wiring Diagram



(The attached diagram shows complete system wiring between PLC modules, drive board, motor, encoder, and lamps.)

4. System Operation

The PLC is the main controller.

When START is pressed:

1. PLC sets run bit
2. Speed ramps upward gradually
3. Analog voltage sent to DC drive
4. Motor starts rotating

When STOP is pressed:

1. PLC ramps speed downward
2. Output voltage reduces gradually
3. Motor stops smoothly

The encoder continuously feeds pulse signals back to PLC for speed monitoring.

5. Analog Output Scaling (0–10 V and 0–4000 Units)

The analog module uses PLC engineering units.

PLC Units Output Voltage

0	0 V
1000	2.5 V
2000	5.0 V
3000	7.5 V
4000	10.0 V

Formula Used

$$Voltage = \frac{Speed\ Units}{4000} \times 10$$

6. Speed Setpoint Scaling

The operator enters desired speed in RPM.

PLC converts RPM to analog units using:

$$AT_SETPOINT_SPEED = \frac{AT_SET_RPM \times 4000}{27.6}$$

Where:

- 27.6 RPM = maximum tested speed
- 4000 = full analog scale

7. Actual RPM Measurement Using Encoder

The encoder generates pulses while motor rotates.

PLC counter counts pulses every 1 second.

Formula Used

$$RPM = \frac{Pulse\ Count}{200} \times 60$$

Where:

- 200 = pulses per revolution
- 60 converts rev/sec to RPM

8. Ramp Up and Ramp Down Logic

Smooth ramping prevents sudden starts and stops.

The operator enters:

- Ramp Up Time (sec)
- Ramp Down Time (sec)

Ramp Up Formula

$$AT_RAMP_STEP_UP = \frac{400}{AT_ACC_TIME}$$

Ramp Down Formula

$$AT_RAMP_STEP_DOWN = \frac{400}{AT_DEC_TIME}$$

9. Ladder Logic Summary

The PLC ladder logic included:

Rung 0

Start/Stop seal-in circuit.

Rung 1–2

Speed setpoint conversion.

Rung 5–9

Ramp up/down calculations.

Rung 12–14

Encoder counting and RPM calculation.

Rung 15–16

Analog output voltage generation.

Rung 17–19

Motor status lights:

- Running
- Stopped
- Accelerating
- Decelerating

10. HMI Interface

The HMI provides local control.

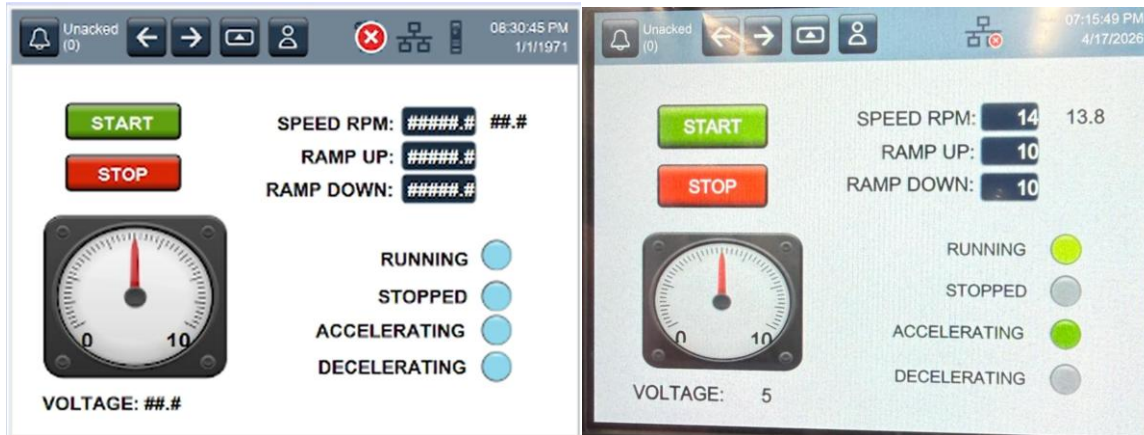
Functions Included

- START button
- STOP button
- Speed setpoint entry
- Ramp Up entry

- Ramp Down entry
- Actual RPM display
- Voltage gauge
- Status lamps

Tested Result

- Set Speed = 14 RPM
- Actual Speed = 13.8 RPM
- Voltage = 5 V



11. SCADA Interface

The SCADA system provides supervisory control and real-time monitoring.

Features

- Live process values
- Start / Stop commands
- Historical trends
- Motor status
- Operator adjustments



12. Real-Time Graph Analysis

The graph plots:

- **Green = Actual Motor RPM**
- **Blue = Output Voltage**

Observations

During Start

Green line rises quickly showing speed increase.

During Stop

Speed drops to zero.

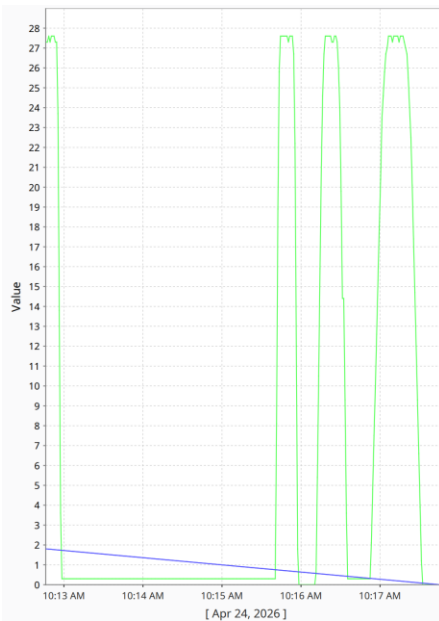
Repeated Operation

Multiple start-stop cycles were observed, proving stable repeatable operation.

Voltage Trend

Blue line shows analog command changes sent from PLC.

This confirms communication between PLC and SCADA.



13. Observations

Overshoot

Speed briefly exceeded setpoint during acceleration.

Reasons:

- Motor inertia
- Fast ramp rate
- Delay in feedback update

Undershoot

Speed slightly below setpoint.

Reasons:

- Load friction
- Voltage drop
- Sampling delay

Stable Operation

Final steady speed remained close to target:

- Set = 14 RPM
- Actual = 13.8 RPM

Only small error observed.

14. Challenges Faced

Our group experienced:

- Encoder pulse setup
- Analog calibration
- HMI communication
- Trend graph configuration
- Matching commanded speed to actual speed

These were solved through repeated testing and tuning.

15. Conclusion

This project successfully demonstrated a full PLC-based DC motor control system using industrial automation hardware and software.

The motor speed was controlled using **0–10 V analog output scaled from 0–4000 PLC units**. Encoder feedback allowed real-time RPM monitoring. Adjustable ramp times provided smooth acceleration and deceleration.

The system operated reliably and accurately while integrating PLC programming, electrical wiring, HMI control, SCADA monitoring, and feedback measurement.

16. Recommendations

Future improvements may include:

- PID closed-loop control
- Reverse direction control
- Overload protection
- RPM filtering
- Auto tuning of ramps